

1 Abstract

This study evaluates the temporal stability of High-Performance Liquid Chromatography (HPLC) detector signals across 11 distinct chromatographic runs. The primary objective was to identify gradual drift patterns indicative of detector instability, lamp degradation, or system fouling, while distinguishing these trends from inherent process variability. Due to the dataset structure containing 1.08 million fraction collection records, analysis was restricted to the 11 unique runs, aggregating data by chromatography stage order (2–20) to establish a chronological timeline. The methodology employed linear regression slope calculation, rolling mean standard deviation analysis, Shewhart control charting, and detrended residual analysis. Key findings indicate that while UV absorbance signals at 280nm and 260nm exhibit consistent linear trends across stages, suggesting stable detector performance, minor non-linearities were observed in the residual variance. System stability metrics, specifically conductivity, showed no significant correlation with UV drift, reinforcing that observed trends are likely intrinsic to the detector optics rather than sample preparation errors. The absence of critical control variables such as injection volume and concentration limits the ability to attribute drift to flow rate instability; however, the current analysis confirms that the optical system maintained integrity throughout the 19-stage timeline, with no evidence of severe lamp degradation or column fouling requiring immediate corrective maintenance.

2 Introduction

The integrity of chromatographic data relies heavily on the stability of detector signals and system parameters over time. In this workflow, the focus area is the systematic evaluation of temporal trends in HPLC detector signals, specifically UV absorbance at 280nm and 260nm, alongside system stability metrics like conductivity and pressure. The primary motivation for this analysis is to isolate gradual drift patterns from inherent process variability and noise. Such drift can be indicative of critical issues such as detector instability, lamp degradation, or column fouling, all of which can compromise product quality and data integrity. The dataset under review comprises 1.08 million records representing fraction collections, but is structured around only 11 unique run IDs. This structure necessitates an analysis approach that utilizes ‘chromatography_stage_order’ as the temporal axis rather than ‘run_no’, focusing on the high-completeness optical signals while acknowledging the significant absence of control variables like injection volume and concentration. These missing variables prevent the validation of whether drift is caused by flow rate instability or sample concentration changes; therefore, the analysis relies primarily on internal detector metrics and fully populated physical parameters. Furthermore, the presence of negative values in continuous variables suggests potential data entry errors or baseline artifacts that required specific handling, such as flagging rather than clipping, to preserve information regarding baseline stability. The objective of this report is to detail the methodology used to detect trends and the main results regarding the stability of the HPLC system across the defined runs.

3 Methodology

The data analysis workflow was executed in three distinct phases to ensure robust trend detection and stability assessment. The first phase involved data preprocessing and temporal aggregation. Rows were filtered to include only stages between 2 and 20, and data was grouped by ‘run_no’ to prevent conflating run variability with stage drift. Within each run, data was aggregated by ‘chromatography_stage_order’, calculating mean and standard deviation for the variables of interest: ‘UV_1_280_mAU’, ‘UV_2_260_mAU’, and ‘Cond_mS.cm’. Approximately 76% of rows contained missing ‘chromatography_stage_order’ values and were dropped prior to grouping. Negative signal values were flagged as outliers in the log to preserve baseline artifact information rather than being clipped to zero. The second phase focused on trend detection and control charting. Linear regression was performed on the aggregated data to calculate slopes against ‘chromatography_stage_order’. Rolling means and standard deviations were calculated using a window size of 5 to capture short-term instabilities across the 19-stage timeline. Shewhart control charts with 3-sigma limits were generated for both UV channels, and ‘Cond_mS.cm’ was used to verify pressure stability. The third phase involved correlation and residual analysis. A correlation matrix was calculated between the

UV signals and conductivity for each run. Detrended residual analysis was conducted by subtracting the linear trend from the original UV signals, and the variance of these residuals was calculated for each stage. This allowed for the determination of whether drift was linear (consistent slope) or non-linear (high residual variance). The analysis was constrained by the missing metadata for injection volume and concentration, limiting the scope to internal detector metrics and conductivity.

4 Results and Discussion

The analysis of the HPLC data revealed distinct patterns in detector stability across the 11 runs. As illustrated in Figure 1, the line plot of stage order versus mean UV absorbance demonstrates that the signals for both 280nm and 260nm channels remain within the shaded 3-sigma control limits throughout the 19-stage timeline. The data points exhibit a consistent upward or downward trajectory depending on the specific run, but none of the aggregated means exceed the upper or lower control limits, indicating that the detector signals are statistically stable and not subject to extreme outliers or sudden spikes. This stability suggests that the detector optics and light source are functioning correctly without significant lamp degradation or electronic noise interference. Figure 2 provides further insight through the bar chart of regression slopes, which compares the rate of change for both UV channels across the stages. The slopes are relatively uniform across the stages, supporting the conclusion that the drift, if present, is linear rather than non-linear. The absence of stages with significantly positive or negative slopes that deviate from the general trend reinforces the finding of consistent detector performance. In the discussion of these results, it is important to note that while the visual trends are stable, the correlation analysis (not depicted in the figures but derived from the same dataset) indicated no strong correlation between the UV drift and conductivity ('Cond_mS.cm'). This dissociation suggests that the observed trends are intrinsic to the optical system and not driven by changes in system pressure or flow dynamics. The handling of negative values, flagged as outliers in the preprocessing phase, did not introduce significant bias into the regression models, as the flagged values were excluded from the slope calculations. Overall, the combination of control chart adherence and consistent regression slopes confirms that the HPLC system maintained high data integrity throughout the analyzed runs, with no immediate evidence of requiring corrective maintenance for the detector or optical components.

5 Conclusion

The systematic evaluation of the HPLC detector signals and system stability metrics across 11 distinct runs confirms the robustness of the analytical system. The methodology, which integrated linear regression, rolling statistics, and Shewhart control charting, successfully isolated temporal trends from process noise. The main results indicate that the UV absorbance signals at 280nm and 260nm exhibit stable, linear trends with no evidence of significant drift exceeding 3-sigma limits. The consistency of regression slopes across the chromatography stages further supports the conclusion that the detector is operating within normal parameters, ruling out severe lamp degradation or column fouling as primary causes of signal variation. The lack of correlation between UV signals and conductivity metrics reinforces that the observed stability is independent of system pressure fluctuations. However, the analysis is constrained by the absence of critical control variables such as injection volume and concentration, which limits the ability to definitively rule out sample preparation errors as a source of variability. Despite this limitation, the current evidence strongly suggests that the optical system and detector electronics have maintained integrity throughout the 19-stage timeline. No corrective maintenance actions are required based on the current data, and the system is deemed suitable for continued operation. Future analyses should prioritize the collection of missing metadata to enable a more comprehensive assessment of the relationship between sample preparation and detector stability.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files